TEACHING PLAN

1. Course Name	Programming Practice
2. Course Code	EEE312
3. Credit Value	2
4. Academic Session	202209
5. Lecturer(s)	Hamid Tahaei
6. Tutor(s) (if any)	NIL
7. Course Learning Outcomes (CLO)	At the end of the course, students will be able to: 1. Design program using advanced algorithms and techniques. 2. Use modern programming techniques to solve scientific problems.
8. Sequence of planned lessons	**Topics**
Week 1	Topic 1 Advanced programming algorithms and techniques - Programming for complex number computation
Week 2	Topic 1 Advanced programming algorithms and techniques - Numerical solution for calculating determinant of a matrix
Week 3	Topic 1 Advanced programming algorithms and techniques - Numerical solution for calculating inverse of a matrix using Gauss-Jordan method
Week 4	Topic 1 Advanced programming algorithms and techniques - Eigenvalue and eigenvector using Power method
Week 5	Topic 2 Systems of linear and nonlinear equations - Solving linear equations using Gaussian elimination
Week 6	Topic 2 Systems of linear and nonlinear equations - Solving linear equations using LU decomposition method
Week 7	Topic 2 Systems of linear and nonlinear equations - Solving nonlinear equations using Gauss-Seidel method
Week 8	Topic 2 Systems of linear and nonlinear equations - Solving nonlinear equations using bisection method
Week 9	Topic 2 Systems of linear and nonlinear equations - Solving nonlinear equations using Newton-Raphson method
Week 10	Topic 2 Systems of linear and nonlinear equations - Solving nonlinear equations using fixed-point iteration and secant method
Week 11	Topic 3 Numerical differentiation and integration - Numerical differentiation using the forward, backward, centre difference rules for first and second derivatives.
Week 12	Topic 3 Numerical differentiation and integration - Numerical integration using the trapezoidal rule and the Simpson's formula.
Week 13	Topic 4 Interpolation and Approximation - Interpolation using Lagrange polynomial
Week 14	Topic 4 Interpolation and Approximation - Approximation using the Least-Squares method
Week 15	• Quiz • Review of Assignment 2 solutions

9. Required Materials (including equipment & technology)

No.	List of Practical Activities
1.	**Activities:** - Programming for complex number computation **Materials/ Software used:** • CodeBlocks IDE
2.	**Activities:** - Numerical solution for calculating determinant of a matrix **Materials/ Software used:** • CodeBlocks IDE
3.	**Activities:** - Numerical solution for calculating inverse of a matrix using Gauss-Jordan method **Materials/ Software used:** • CodeBlocks IDE
4.	**Activities:** - Eigenvalue and eigenvector using Power method **Materials/ Software used:** • CodeBlocks IDE
5.	**Activities:** - Solving linear equations using Gaussian elimination **Materials/ Software used:** • CodeBlocks IDE
6.	**Activities:** - Solving linear equations using LU decomposition method **Materials/ Software used:** • CodeBlocks IDE
7.	**Activities:** - Solving nonlinear equations using Gauss-Seidel method **Materials/ Software used:** • CodeBlocks IDE
8.	**Activities:** - Solving nonlinear equations using bisection method **Materials/ Software used:** • CodeBlocks IDE
9	**Activities:** - Solving nonlinear equations using Newton-Raphson method **Materials/ Software used:** • CodeBlocks IDE

	10.	**Activities:** - Solving nonlinear equations using fixed-point iteration and secant method **Materials/ Software used:** • CodeBlocks IDE
	11.	**Activities:** - Numerical differentiation using the forward, backward, centre difference rules for first and second derivatives. **Materials/ Software used:** • CodeBlocks IDE
	12.	**Activities:** - Numerical integration using the trapezoidal rule and the Simpson's formula. **Materials/ Software used:** • CodeBlocks IDE
	13.	**Activities:** - Interpolation using Lagrange polynomial **Materials/ Software used:** • CodeBlocks IDE
	14.	**Activities:** - Approximation using the Least-Squares method **Materials/ Software used:** • CodeBlocks IDE

10. Assessments

Continuous Assessment	**50%**
• Assignment	30%
• Quiz	20%
Final Assessment	**50%**
• Final Project Report	50%
Total	**100**%

11. Assessments that align to Course Learning Outcomes (CLO)

Assessment	CLO 1	CLO2	CLO3	CLO4	CLO5
Continuous Assessment					
• Assignment	√	√			
• Quiz	√				
Final Assessment					
• Final Project	√	√			

12. Main Reference(s)

- S. Nomura, C Programming and Numerical Analysis: An Introduction, Morgan & Claypool Publishers 2018

13. Additional Reference(s)	• S. Salleh, A. Zomaya, S. Bakar, Computing for Numerical Methods Using Visual C++, Wiley-Interscience 2007.